

Dear Editors of the *Journal of Economics and Development* Special Issue on “Sustainable Growth and Welfare in the Age of Artificial Intelligence”,

Please consider our manuscript, “The Pre-AI Digital Divide and Firm Productivity across Asia’s Institutional Regimes: Firm-Level Evidence from 50 Economies”, for the special issue.

The special issue asks how AI will shape development and welfare, particularly in developing and emerging economies. Our paper addresses a precondition that the AI debate often presumes but rarely measures: the basic digital substrate on which any AI dividend must build. Using harmonised World Bank Enterprise Survey micro-data for 88,869 firms across 50 Asian and Pacific economies, we map the firm-level digital divide across six institutional regimes and estimate the productivity dividend of basic digital adoption.

We report three findings of direct relevance to the issue’s themes (AI and development strategy; digital divides and adoption barriers; AI, inequality, and welfare). First, a pronounced firm-level digital divide: web adoption falls from 69% in the strongest-institution regime to roughly 41% in the emerging and Pacific-SIDS regimes, and barely half of the region’s formal-sector firms have crossed even this basic threshold. Second, the productivity dividend from crossing it is positive and significant in every regime (+0.18 to +0.48 SD), and roughly double the level effect of deeper technological capability. Third, the economies furthest behind still earn a significant dividend from closing the gap, yet are least equipped to absorb AI — implying that, unless the pre-AI digital divide is closed, AI is more likely to widen than narrow cross-economy inequality. We draw out a sequencing logic for development policy: digital substrate first, then AI.

The manuscript is approximately 4,500 words, has not been published, and is not under consideration elsewhere. All quantitative results are reproducible from the accompanying scripts; we include an AI-use disclosure. We believe the paper’s firm-level, cross-institutional evidence on the digital substrate of the AI era fits the special issue’s developmental focus.

Sincerely, The Authors